Nikita L. Polomoshnov

Computational drug design · machine learning · Moscow, Russia
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Research interests. Physics-informed, interpretable machine learning: computing what physics or geometry fixes and learning only the residual — and measuring where a composed pipeline silently misleads, through unidentifiable quantities, uncalibrated uncertainty, and metrics whose winner changes with an unstated convention. Two of my three manuscripts report pre-registered negative results as their central finding.

EDUCATION

Lomonosov Moscow State University — Faculty of Bioengineering & Bioinformatics expected 2029

Specialist Degree (Специалист) — 6-year programme; Master's-equivalent level, grants access to doctoral study. GPA 3.8/4.0.

- Second-prize diploma, XI and XII International Biological Universiade, Lomonosov MSU (2023 and 2024).

EXPERIENCE

Yandex — Machine Learning Intern October 2025 – February 2026

- Image-quality assessment: ranking and quality-classification models. Tag suggestion and SQL data preparation. Python, PyTorch, SQL.

Institute of Biomedical Chemistry (IBMC), Moscow — Researcher 2023 – present

- MetaTox on the way2drug platform — metabolite-prediction service going into production; Application Note in preparation. My contribution: a new prediction model and a new bank of transformation rules.
- Enantiomer screening campaign on molecular cages with the Novosibirsk Institute of Organic Chemistry (NIOCH SB RAS): docking and target scoring, with an operational report separating enantiomers on one target.

PUBLICATIONS & MANUSCRIPTS

*Author order reflects contribution; * denotes sole authorship.*

Ground Truth That Does Not Ground: substituting reference σ-profiles degrades COSMO-SAC solubility prediction

N. L. Polomoshnov, A. V. Rudik (IBMC) · under review, JCIM

- A differentiable COSMO-SAC layer lets a graph network train through a fixed thermodynamic map; an exact decomposition separates the map's own misspecification from what its inputs fail to carry. Substituting the tabulated reference profile at inference degrades prediction at every seed, while supervising against it during training carries no established sign. doctawho42.github.io/tgnn-solv

What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network *

N. L. Polomoshnov — sole author · in prep., JCTC

- The per-edge BAR uncertainty, calibrated against replicate scatter on 1145 public OpenFE binding edges, becomes the null of a cycle-closure goodness-of-fit test; a gradient–cycle decomposition fixes exactly what such a test can and cannot see. Public analysis of the benchmark in OpenFE issues #331 and #332.

Machine-learning evaluation methodology *

N. L. Polomoshnov — sole author · under review, ICLR 2027

- Title and details withheld during double-blind review.

SELECTED OPEN-SOURCE & COMMUNITY

- OpenFreeEnergy / IndustryBenchmarks2024 — issue #331 (per-edge observability audit of the public networks, with reproducible tables and a 33-check self-test) and issue #332 (ligand-name collision in the cdk8 set). Public contributions to the 15-company industry FEP benchmark.

TECHNICAL SKILLS

Machine learning: PyTorch; graph neural networks; E(3)-equivariant architectures; deep ensembles; calibration & uncertainty quantification; conformal prediction; GFlowNets.

Simulation & cheminformatics: molecular dynamics & alchemical free-energy calculations (OpenFE / OpenFF, OpenMM); MBAR/BAR (pymbar); COSMO-SAC / COSMO-RS; RDKit; docking.

Bioinformatics: sequence analysis and homology search (BLAST); sequence and structure databases; NGS data processing; structural bioinformatics — PyMOL, molecular dynamics (GROMACS); classical sequence- and structure-analysis algorithms.

Statistics: M-estimation & sandwich variance; bootstrap (paired, clustered); multiplicity control; pre-registration; negative-control design.

Engineering: reproducible pipelines; artifact / DOI release; Git, Docker, SQL.

Languages: Russian (native). English (professional working proficiency; formally qualified in scientific-text review and translation).